Strict weighted mean-convexity does not control the Neumann gap

Autonomous partial-result packet for arXiv:1310.2526

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Abstract

Kolesnikov and Milman ask whether Veysseire's harmonic-mean curvature spectral-gap estimate has analogues in their Dirichlet and generalized mean-convex Neumann regimes. We show that the pure Neumann estimate cannot hold under generalized mean-convexity alone. In every dimension at least three there are smooth bounded Euclidean dumbbells with strictly positive weighted mean curvature and constant positive Bakry–Émery curvature, but whose weighted Neumann gap tends to zero. Hence any viable analogue of their Case (3) must retain a boundary-trace correction (or impose the stronger second-fundamental-form condition). The boundary-corrected Case (3) and the Dirichlet question are not settled here.

1 Statement

On a smooth weighted domain $(\Omega, g, \mu = e^{-V} d\mathrm{vol})$, write

$$\mathrm{Ric}_\mu = \mathrm{Ric}_g + \nabla^2 V, \qquad H_\mu = H - \langle \nabla V, \nu \rangle,$$

where ν is the outer unit normal and H is the sum of the principal curvatures. Let $\lambda_1^N(\Omega, \mu)$ be the first nonzero eigenvalue of the weighted Neumann Laplacian, equivalently the best constant in

$$\lambda_1^N \operatorname{Var}_\mu(f) \le \int_\Omega |\nabla f|^2 \, d\mu.$$

Theorem 1 (mean-convex dumbbell obstruction)**.** *For every $n \ge 3$ there are constants $\kappa, c, R > 0$ and a family of smooth, bounded, connected domains $\Omega_\varepsilon \subset B_R(0) \subset \mathbb{R}^n$, $0 < \varepsilon < \varepsilon_0$, such that for*

$$d\mu_\varepsilon = e^{-\kappa |x|^2/2} \, dx$$

one has

$$\begin{aligned} \mathrm{Ric}_{\mu_\varepsilon} &= \kappa g_{\mathrm{Eucl}} && \text{on } \Omega_\varepsilon, \\ H_{\mu_\varepsilon} &\ge c && \text{on } \partial\Omega_\varepsilon, \end{aligned}$$

but

$$\lambda_1^N(\Omega_\varepsilon, \mu_\varepsilon) \le C_n \varepsilon^{n-1} \longrightarrow 0.$$

Consequently the Veysseire-form estimate

$$\lambda_1^N \ge \left(\frac{1}{\mu(\Omega)} \int_\Omega \frac{1}{\rho} \, d\mu \right)^{-1} \tag{1}$$

is false if local convexity is weakened to strict generalized mean-convexity, even when $\mathrm{Ric}_\mu = \rho g$ with ρ constant.

2 Geometry of the domains

We use the following standard mean-convex connected-sum construction, and include the relevant quantitative features.

Lemma 1 (uniform mean-convex dumbbells)**.** *For every $n \geq 3$ there are $R, h_0 > 0$ and smooth domains $\Omega_\varepsilon \subset B_R(0)$, symmetric under $(x_1, x') \mapsto (-x_1, x')$, with the following properties:*

1. *Ω_ε contains two fixed positive-volume lobes, exchanged by the symmetry;*
2. *the lobes are joined by the straight cylinder $(-1,1) \times B_\varepsilon^{n-1}$;*
3. *$H_{\partial\Omega_\varepsilon} \geq h_0$ everywhere.*

Proof. Start with two fixed round balls and join them by a rotationally symmetric tube. If the tube boundary is written as

$$(s, r_\varepsilon(s)\theta), \qquad \theta \in S^{n-2},$$

then its outward mean curvature is

$$H = -\frac{r_\varepsilon''}{(1+(r_\varepsilon')^2)^{3/2}} + \frac{n-2}{r_\varepsilon\sqrt{1+(r_\varepsilon')^2}}. \tag{2}$$

Take $r_\varepsilon = \varepsilon$ on $[-1,1]$. Immediately outside this interval, increase the radius while it is still uniformly small, using a profile with bounded positive second derivative. The second term in (2) then dominates uniformly. Continue with a concave increasing profile, for which both displayed contributions are nonnegative, and attach it to a fixed spherical lobe. The inequalities can be made strict with a margin independent of ε. Reflect the construction at $s = 0$ and smooth the finitely many joins inside those strict inequalities. The central cylinder has $H = (n-2)/\varepsilon$, the spherical pieces have fixed positive mean curvature, and the compact transition pieces retain a uniform lower bound. This proves the lemma. □

3 Weight and Rayleigh quotient

Proof of Theorem 1. Apply Lemma 1. Since every domain lies in $B_R(0)$, choose

$$\kappa = \frac{h_0}{2R}, \qquad V(x) = \frac{\kappa}{2}|x|^2.$$

Euclidean Ricci curvature is zero and $\nabla^2 V = \kappa g$, hence $\mathrm{Ric}_{\mu_\varepsilon} = \kappa g$. On the boundary,

$$H_{\mu_\varepsilon} = H - \kappa\langle x, \nu\rangle \geq h_0 - \kappa|x| \geq \frac{h_0}{2}.$$

Thus the weighted domains are strictly generalized mean-convex, uniformly in ε.

It remains to estimate their Neumann gap. Let f_ε be odd under the reflection in x_1, equal to -1 on the left lobe and 1 on the right lobe, and interpolate linearly along the central cylinder. It may be chosen Lipschitz, with $|\nabla f_\varepsilon| \leq C$ and gradient supported in a fixed-length portion of the cylinder. Symmetry of both the domain and the weight gives $\int f_\varepsilon \, d\mu_\varepsilon = 0$. The two fixed lobes and the uniform bounds on V give

$$\int_{\Omega_\varepsilon} f_\varepsilon^2 \, d\mu_\varepsilon \geq c_0 > 0.$$

The cylinder has cross-sectional volume $\omega_{n-1}\varepsilon^{n-1}$, so

$$\int_{\Omega_\varepsilon} |\nabla f_\varepsilon|^2 \, d\mu_\varepsilon \le C_n \varepsilon^{n-1}.$$

The Neumann Rayleigh principle yields $\lambda_1^N \le C_n \varepsilon^{n-1}$.

Finally, in this example one may take $\rho \equiv \kappa$, and therefore

$$\left(\frac{1}{\mu_\varepsilon(\Omega_\varepsilon)} \int_{\Omega_\varepsilon} \frac{1}{\rho} \, d\mu_\varepsilon \right)^{-1} = \kappa.$$

For small ε, this contradicts (1). □

4 What this does and does not answer

The locally convex theorem in arXiv:1310.2526 uses $\mathrm{II}_{\partial M} \ge 0$. Under Neumann boundary conditions, the Reilly formula contains

$$\int_{\partial M} \mathrm{II}_{\partial M}(\nabla_{\partial M} u, \nabla_{\partial M} u) \, d\mu_{\partial M},$$

whose sign is not controlled by the trace H_μ. The dumbbell construction is the geometric manifestation of that missing tensor control.

Case (3) of Kolesnikov–Milman contains an additional boundary-trace variance weighted by H_μ^{-1}. The test above has different traces on the two lobes, so that term does not disappear; the theorem does not refute such a boundary-corrected Veysseire analogue. Likewise, a Dirichlet function vanishes on the entire boundary, and the same low-energy lobe test is unavailable. Those two formulations remain open in this packet.

References

[1] A. V. Kolesnikov and E. Milman, *Brascamp–Lieb type inequalities on weighted Riemannian manifolds with boundary*, arXiv:1310.2526.

[2] L. Veysseire, *Improved spectral gap bounds on positively curved manifolds*, arXiv:1105.6080.